Mohammad Sarhangzadeh

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Research Interests

- Computer Vision
- Medical Image Analysis
- Generative Modeling
- Uncertainty Quantification
- Motion Analysis
- Topological Learning

Education

Bilkent University Ankara, Türkiye
M.Sc. in Computer Engineering *Sep 2025–Present*
Fully funded M.Sc. program
Supervisor: Dr. Doruk Öner

University of Tabriz Tabriz, Iran
B.Sc. in Computer Engineering *2020–2025*

Publications

Stochastic Nonlinearities Improve Uncertainty Estimation
Mohammad Sarhangzadeh, Amirreza M. Shebly, Doruk Öner
British Machine Vision Conference (BMVC), 2026

Direction-Aware Kinematics-Conditioned Graph Hypernetwork for Human Trajectory Prediction
Amirreza M. Shebly, Mohammad Sarhangzadeh, Doruk Öner
Asian Conference on Computer Vision (ACCV), 2026 ORAL

DINO-Nucleus: Multi-Level Self-Supervised Features for Nucleus Instance Segmentation
Mohammad Sarhangzadeh, Elyar Esmaeilzadeh, Doruk Öner
International Conference on Medical Imaging and Computer-Aided Diagnosis (MICAD), 2026

OMCDE: Opposition-based Multi-Stage Matrix-Based Competitive Differential Evolution
Mohammad Sarhangzadeh, Shahryar Rahnamayan, Doruk Öner
IEEE Congress on Evolutionary Computation (CEC), WCCI 2026 ORAL

Ongoing Research

Topology-Aware Uncertainty Estimation for Structured Segmentation
Studying uncertainty in the connectivity and topology of segmented structures, with an emphasis on detecting structural failures that conventional pixel-level confidence estimates may miss. Investigating uncertainty-aware learning and evaluation for curvilinear structures.

Diffusion-Based Tilt-Series Extrapolation for Limited-Angle Tomography
Investigating diffusion-based generation of unobserved projection views from sparsely sampled tilt series to extend the effective angular range of tomography. Exploring measurement fidelity, cross-view and cycle consistency, Fourier-domain constraints, and preservation of reconstructed 3D structure.

Research Experience

NeuraVision Lab, Bilkent University Ankara, Türkiye
Graduate Researcher *Sep 2025–Present*

- **Stochastic activation methods for uncertainty quantification (BMVC 2026).** Developed a single-model approach that introduces stochasticity through nonlinear activation functions, enabling repeated predictive samples without maintaining multiple separately trained models. Investigated how activation trajectories support uncertainty and calibration, and evaluated the approach on image classification, segmentation, out-of-distribution detection, and input corruption.
- **Direction-aware trajectory forecasting (ACCV 2026).** Investigated how agents' motion and relative direction should influence interaction modeling in crowded environments. Contributed to a kinematics-conditioned graph hypernetwork that adapts message passing to directional relationships, with an emphasis on learning useful interactions for multimodal human-trajectory prediction.
- **Self-supervised medical instance segmentation (MICAD 2026).** Developed DINO-Nucleus, combining hierarchical self-supervised visual representations with multi-level feature fusion for nucleus instance segmentation. Explored parameter-efficient adaptation and complementary training objectives to distinguish closely packed nuclei while limiting the number of trainable parameters.
- **Competitive differential evolution (WCCI 2026).** Developed OMCDE, a matrix-vectorized population optimizer that combines opposition-based initialization, fitness-ranked triplet competition, adaptive mutation and crossover parameters, and a staged shift from exploration to exploitation. Studied numerical optimization behavior on continuous benchmark functions.

Applied Deep Learning Lab, University of Tabriz Tabriz, Iran
Undergraduate Research Assistant *May 2022–Aug 2025*

- **Medical image analysis.** Conducted research on deep learning for volumetric brain-tumor segmentation and pediatric kidney-tumor classification from MRI. Worked with U-Net-based segmentation and VAE-derived latent representations to investigate accurate, computationally efficient medical-image models.
- **Pose-guided sign-language video generation.** Led research and model development for generating sign-language videos conditioned on pose sequences. The initial stage focused on synthesizing video directly from sign-language pose information; subsequent work explored fine-tuning diffusion-based video models and maintaining visual and temporal consistency.

Intellectual Property

Invention Disclosure / Potential Patent Protection *Ongoing*

Amirreza M. Shebly, Mohammad Sarhangzadeh, Doruk Öner

Invention disclosure under institutional evaluation for potential patent protection of research conducted within a TÜBİTAK BİLGEM-funded project.

Teaching Experience

Bilkent University *Teaching Assistant*

Python Programming, Digital Design, Generative Networks

University of Tabriz *Teaching Assistant*

Data Structures, Logical Circuits

Languages

English (IELTS Academic: 7.5) · Turkish (native) · Persian (native)